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Learning an Autonomous Drone Racing Policy via Differentiable Simulation

Master's Thesis

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Abstract

Todo

Keywords Unmanned Aerial Vehicles, Autonomous Drone Racing, Simulation to Reality Transfer, Reinforcement Learning, Domain Randomization

Abbreviations

DR Domain Randomization

Sim2Real Simulation-to-Reality

ADR Autonomous Drone Racing

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■ 1 Introduction

■ 1.1 Background

Give context about the problem

What is ADR?

- Navigate through a sequence of gates as fast as possible within the platform capabilities...

What are the challenges of ADR?

- **Where am I?** → Perception
challenges: limited sensory data, noisy and uncertain partial observability
- **Where do I go?** → Planning
safe efficient path planning
- **How do I get there?** → Control
Handle the non-linear underactuated dynamics to execute the planned motion under actuator limits and model uncertainties

■ 1.2 Related Work

Present the state of the art, how do the different research labs approach this problem

Open Questions

- How to improve sample efficiency
- Generalization to multiple tasks (how to race while avoiding obstacles, how to race against other drones)
- Transferability of the controllers under unmodeled effects

■ 1.3 Structure and Objectives

■ 2 Preliminaries

■ 2.1 Optimal Control Problem

Formulate the optimal control problem, and

■ Pontryagin's Maximum Principle

maybe solve it using a pmm and Pontryagin's Maximum Principle

Since it is impossible to perfectly model the state transition equation, and the state estimation is not perfect, the optimal solution breaks under real world conditions.

■ 2.2 Classical Control approaches

■ Differential Flatness Based Control

■ Model Predictive Control

■ 2.3 Learning-Based approaches

■ Model-Free Reinforcement Learning

■ Model-Based Reinforcement Learning

■ 3 Methodology

This chapter presents the methodology used in this thesis to train the neural network policies. In section 3.1, we address the core elements of the simulation architecture, define the quadrotor model, and explain how the Simulation-to-Reality (Sim2Real) gap is minimized by using Domain Randomization (DR). Furthermore, we introduce the implemented control abstraction levels. Section 3.2, outlines the training objectives used for Autonomous Drone Racing (ADR). Lastly, section 3.3 details how differentiability is leveraged to optimize the policies in order to minimize the training task's cost function.

■ 3.1 Simulation Framework

[Insert cool diagram](#)

■ Quadrotor Model

■ Domain Randomization

■ Control Modes

■ 3.2 Training Tasks

■ 3.3 Differentiable Optimization

■ 4 Results

- 4.1 Evaluation Criteria
- 4.2 Baseline Performance
- 4.3 Ablation Study
- 4.4 Transferability to MRS UAV System
- 4.5 Real World Validation

5 Conclusion

■ 6 Future Directions

■ 7 References

■ A Appendix A